

Dividend timing and the equity index forward: a real-data check on FEZ

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The question

How much does it actually matter if you build an index forward from a single flat trailing dividend yield, rather than from the real, discrete dividend schedule? This is a check using live options data on FEZ (SPDR EURO STOXX 50 ETF), a US-listed, liquid ETF tracking the EURO STOXX 50, pulled on 26 July 2026.

A note on the proxy: Eurex doesn't give away native EURO STOXX 50 index option (OESX) data for free, so this uses FEZ instead. It's a real, liquid, listed options market, not the index itself, and it's USD-denominated and unhedged against a EUR index. Furthermore, FEZ pays out on its own smoothed ETF distribution schedule, which inherently masks the concentrated spring dividend cliff of the native EURO STOXX 50 constituents. Worth saying upfront rather than pretending otherwise.

Method

For each expiry, three forwards get compared:

1. **Market forward** – recovered directly from the chain via put-call parity, $C(K) - P(K) = DF(T)(F - K)$. This is the market's own truth and needs no dividend assumption at all.
2. **Naïve forward** – $F = S_0 e^{(r-q)T}$ using a single flat trailing dividend yield.
3. **Discrete forward** – built from FEZ's actual historical distribution dates and amounts, rolled forward one year, same calendar timing, no growth assumed.

Since FEZ trades American-style, every quote gets converted to a European-equivalent price first (Barone-Adesi-Whaley implied vol, then the European price at that vol) before touching parity or an SSVI fit. Skip this step and early-exercise premium contaminates the recovered forward, and that contamination is indistinguishable from the dividend-timing effect this is meant to measure.

The SSVI slice fit is spread-weighted (inverse-squared bid-ask spread), not plain least squares. A first pass with an unweighted fit let wide, illiquid wing quotes drag the calibration away from the tight ATM quotes, which matters most for anyone actually pricing off this surface.

Expiry	Naive gap (vol-bp)	Discrete gap (vol-bp)	SSVI arb-free?
0.32y	-245.8	+29.9	No (phi pinned at bound)
1.49y	-378.6	+21.0	Yes (though rho pinned near its bound too)

Table 1: Forward mispricing versus the market-implied forward, expressed as an equivalent vol error at the ATM strike.

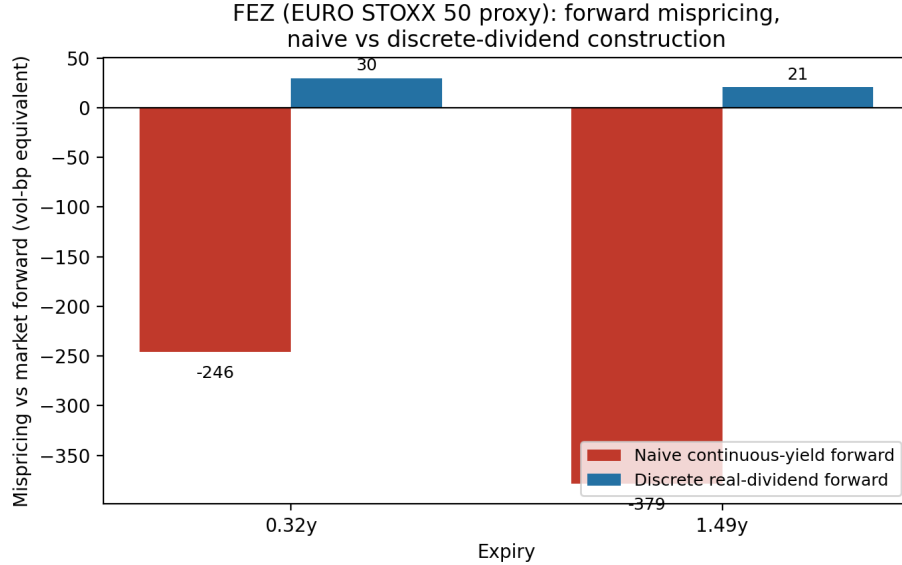


Figure 1: Naive vs discrete-dividend forward mispricing, by expiry.

Results

The naive forward is off by the equivalent of 245 to 380 vol-bp. Swap in the real discrete dividend schedule and that drops to 21 to 30 bp, roughly a tenfold reduction, built entirely from real dividend cash amounts and real dates.

Spread-weighting made almost no difference at 0.32y (367 bp either way) but pulled the 1.49y fit from 234 bp down to 191 bp. Still not tight by professional market-making standards, but a real improvement, and it tells you something: at 1.49y the wide wing quotes genuinely were dragging the fit, whereas at 0.32y the fit quality problem sits somewhere else entirely (see below).

Things worth flagging rather than glossing over

A few honest caveats, because the point of this exercise was to actually understand the data, not just produce a clean-looking number.

The 0.32y slice failed its own no-arbitrage check. Phi got pinned at the calibration's upper bound and the fit RMSE stayed large even after weighting. I'd lean on the 1.49y result as the main finding and treat 0.32y as directionally consistent but not something I'd defend to the decimal place.

The 1.49y slice's rho also landed right on its own bound (-0.999), even though it technically passed the arbitrage check. That's not nothing, it suggests the optimiser wants to go

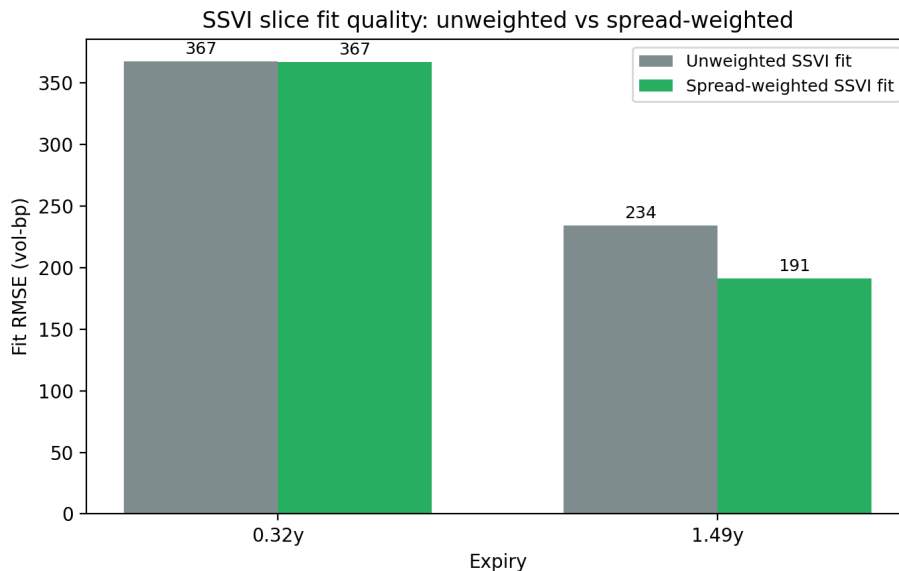


Figure 2: SSVI fit RMSE, unweighted vs spread-weighted, by expiry.

further than the bound allows, and a genuinely converged, well-identified fit shouldn't sit exactly on a boundary. Worth treating the reported ρ and ϕ as directionally right rather than precisely pinned down.

Rate consistency turned out to be a bigger deal than expected. The rate implied by the parity regression itself came out at 0.16% and 2.88% for the two expiries, versus the 3.8% external T-bill proxy used elsewhere in the pipeline, a gap of -364 and -91 vol-bp respectively. That's a genuinely large discrepancy, and it means the parity regression is absorbing the ETF's stock borrow cost into the implied rate. To fix this and trust the naive-gap number to the decimal place, the next step would be applying an external SOFR/OIS curve for the risk-free rate and isolating the residual purely as the implied borrow fee.

Only 2 of the original 5 expiries survived at all, the rest dropped for too few two-sided quotes or quotes that failed to clear intrinsic value once the American-exercise correction went in. That's real thinness in real data (FEZ is a moderately-liquid ETF, not a professionally market-made index book), not a filtering mistake.

The American-exercise premium was small but real, under 3 vol-bp even at 1.49y. An order of magnitude smaller than the dividend-timing effect, but skipping the correction entirely would have baked that premium into the parity forward regardless.

Takeaway

Dividend timing is the biggest lever in this comparison by a wide margin, bigger than the American-exercise correction, and possibly entangled with a borrow-cost effect that's worth isolating properly before leaning on the exact numbers. None of that changes the direction of the result. A flat trailing yield gets the forward wrong by hundreds of vol-bp; a real, dated dividend schedule gets you most of the way back. Worth remembering for anything priced off the same forward: skew, term structure, all of it.